Variance

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For a single variate X having a distribution $P(x)$ with *known* population mean μ, the population variance $\mathrm{var}(X)$, commonly also written σ^2, is defined as

$$\sigma^2 \equiv \left\langle (X-\mu)^2 \right\rangle, \tag{1}$$

where μ is the population mean and $\langle X \rangle$ denotes the expectation value of X. For a discrete distribution with N possible values of x_i, the population variance is therefore

$$\sigma^2 = \sum_{i=1}^{N} P(x_i)\,(x_i-\mu)^2, \tag{2}$$

whereas for a continuous distribution, it is given by

$$\sigma^2 = \int P(x)\,(x-\mu)^2\,dx. \tag{3}$$

The variance is therefore equal to the second central moment μ_2.

Note that some care is needed in interpreting σ^2 as a variance, since the symbol σ is also commonly used as a parameter related to but not equivalent to the square root of the variance, for example in the log normal distribution, Maxwell distribution, and Rayleigh distribution.

If the underlying distribution is not known, then the sample variance may be computed as

$$s_N^2 \equiv \frac{1}{N}\sum_{i=1}^{N}(x_i-\bar{x})^2, \tag{4}$$

where $\bar{x}$ is the sample mean.

Note that the sample variance s_N^2 defined above is *not* an unbiased estimator for the population variance σ^2. In order to obtain an unbiased estimator for σ^2, it is necessary to instead define a "bias-corrected sample variance"

$$s_{N-1}^2 \equiv \frac{1}{N-1}\sum_{i=1}^{N}(x_i-\bar{x})^2. \tag{5}$$

The distinction between s_N^2 and s_{N-1}^2 is a common source of confusion, and extreme care should be exercised when consulting the literature to determine which convention is in use, especially

since the uninformative notation s is commonly used for both. The bias-corrected sample variance s_{N-1}^2 for a list of data is implemented as Variance[*list*].

The square root of the variance is known as the standard deviation.

The reason that s_N^2 gives a biased estimator of the population variance is that two free parameters μ and σ^2 are actually being estimated from the data itself. In such cases, it is appropriate to use a Student's *t*-distribution instead of a normal distribution as a model since, very loosely speaking, Student's *t*-distribution is the "best" that can be done without knowing σ^2.

Formally, in order to estimate the population variance σ^2 from a sample of n elements with a priori *unknown* mean (i.e., the mean is estimated from the sample itself), we need an unbiased estimator for σ^2. This is given by the *k*-statistic $k_2 = \hat{\sigma}^2$, where

$$k_2 = \frac{N}{N-1} m_2 \tag{6}$$

and $m_2 \equiv s_N^2$ is the sample variance uncorrected for bias.

It turns out that the quantity $N s_N^2 / \sigma^2$ has a chi-squared distribution.

For set of data X, the variance of the data obtained by a linear transformation is given by

$$\begin{aligned}
\operatorname{var}(a X + b) &= \left\langle [(a X + b) - \langle a X + b \rangle]^2 \right\rangle && (7)\\
&= \left\langle (a X + b - a \langle X \rangle - b)^2 \right\rangle && (8)\\
&= \left\langle (a X - a \mu)^2 \right\rangle && (9)\\
&= \left\langle a^2 (X - \mu)^2 \right\rangle && (10)\\
&= a^2 \left\langle (X - \mu)^2 \right\rangle && (11)\\
&= a^2 \operatorname{var}(X) && (12)
\end{aligned}$$

For multiple variables, the variance is given using the definition of covariance,

$$\begin{aligned}
\operatorname{var}\left(\sum_{i=1}^n X_i\right) &= \operatorname{cov}\left(\sum_{i=1}^n X_i, \sum_{j=1}^n X_j\right) && (13)\\
&= \sum_{i=1}^n \sum_{j=1}^n \operatorname{cov}(X_i, X_j) && (14)\\
&= \sum_{i=1}^n \sum_{\substack{j=1 \\ j=i}}^n \operatorname{cov}(X_i, X_j) + \sum_{i=1}^n \sum_{\substack{j=1 \\ j\neq i}}^n \operatorname{cov}(X_i, X_j) && (15)\\
&= \sum_{i=1}^n \operatorname{cov}(X_i, X_i) + \sum_{i=1}^n \sum_{\substack{j=1 \\ j\neq i}}^n \operatorname{cov}(X_i, X_j) && (16)\\
&= \sum_{i=1}^n \operatorname{var}(X_i) + 2 \sum_{i=1}^n \sum_{j=i+1}^n \operatorname{cov}(X_i, X_j). && (17)
\end{aligned}$$

A linear sum has a similar form:

$$\operatorname{var}\left(\sum_{i=1}^{n} a_i X_i\right) = \operatorname{cov}\left(\sum_{i=1}^{n} a_i X_i, \sum_{j=1}^{n} a_j X_j\right) \tag{18}$$

$$= \sum_{i=1}^{n} \sum_{j=1}^{n} a_i a_j \operatorname{cov}(X_i, X_j) \tag{19}$$

$$= \sum_{i=1}^{n} a_i^2 \operatorname{var}(X_i) + 2 \sum_{i=1}^{n} \sum_{j=i+1}^{n} a_i a_j \operatorname{cov}(X_i, X_j). \tag{20}$$

These equations can be expressed using the covariance matrix.

SEE ALSO

Central Moment, Charlier's Check, Covariance, Covariance Matrix, Error Propagation, *k*-Statistic, Mean, Moment, Raw Moment, Sample Variance, Sample Variance Computation, Sample Variance Distribution, Sigma, Standard Error, Statistical Correlation

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WolframAlpha variance {21.3, 38.4, 12.7, 41.6} =

More things to try: = variance {21.3, 38.4, 12.7, 41.6} = variance of uniform distribution = variance (1,2,3)

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Papoulis, A. *Probability, Random Variables, and Stochastic Processes, 2nd ed.* New York: McGraw-Hill, pp. 144-145, 1984.

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Variance

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SUBJECT CLASSIFICATIONS

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Covariance

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Covariance provides a measure of the strength of the correlation between two or more sets of random variates. The covariance for two random variates X and Y, each with sample size N, is defined by the expectation value

$$\operatorname{cov}(X, Y) = \langle (X - \mu_X)(Y - \mu_Y) \rangle \tag{1}$$

$$= \langle X\, Y \rangle - \mu_X\, \mu_y \tag{2}$$

where $\mu_x = \langle X \rangle$ and $\mu_y = \langle Y \rangle$ are the respective means, which can be written out explicitly as

$$\operatorname{cov}(X, Y) = \sum_{i=1}^{N} \frac{(x_i - \bar{x})(y_i - \bar{y})}{N}. \tag{3}$$

For uncorrelated variates,

$$\operatorname{cov}(X, Y) = \langle X\, Y \rangle - \mu_X\, \mu_Y = \langle X \rangle \langle Y \rangle - \mu_X\, \mu_Y = 0, \tag{4}$$

so the covariance is zero. However, if the variables are correlated in some way, then their covariance will be nonzero. In fact, if $\operatorname{cov}(X, Y) > 0$, then Y tends to increase as X increases, and if $\operatorname{cov}(X, Y) < 0$, then Y tends to decrease as X increases. Note that while statistically independent variables are always uncorrelated, the converse is not necessarily true.

In the special case of $Y = X$,

$$\operatorname{cov}(X, X) = \left\langle X^2 \right\rangle - \langle X \rangle^2 \tag{5}$$

$$= \sigma_X^2, \tag{6}$$

so the covariance reduces to the usual variance $\sigma_X^2 = \operatorname{var}(X)$. This motivates the use of the symbol $\sigma_{X\,Y} = \operatorname{cov}(X, Y)$, which then provides a consistent way of denoting the variance as $\sigma_{X\,X} = \sigma_X^2$, where σ_X is the standard deviation.

The derived quantity

$$\operatorname{cor}(X, Y) = \frac{\operatorname{cov}(X, Y)}{\sigma_X\, \sigma_Y} \tag{7}$$

$$= \frac{\sigma_{X\,Y}}{\sqrt{\sigma_{X\,X}\, \sigma_{Y\,Y}}}, \tag{8}$$

is called statistical correlation of X and Y.

The covariance is especially useful when looking at the variance of the sum of two random variates, since

$$\mathrm{var}\,(X+Y)=\mathrm{var}\,(X)+\mathrm{var}\,(Y)+2\,\mathrm{cov}\,(X,\,Y). \tag{9}$$

The covariance is symmetric by definition since

$$\mathrm{cov}\,(X,\,Y)=\mathrm{cov}\,(Y,\,X). \tag{10}$$

Given n random variates denoted $X_1, ..., X_n$, the covariance $\sigma_{i\,j}\equiv\mathrm{cov}\left(X_i,\,X_j\right)$ of X_i and X_j is defined by

$$\mathrm{cov}\left(X_i,\,X_j\right)=\left\langle\left(X_i-\mu_i\right)\left(X_j-\mu_j\right)\right\rangle \tag{11}$$

$$=\left\langle X_i\,X_j\right\rangle-\mu_i\,\mu_j, \tag{12}$$

where $\mu_i=\langle X_i\rangle$ and $\mu_j=\left\langle X_j\right\rangle$ are the means of X_i and X_j, respectively. The matrix $\left(V_{i\,j}\right)$ of the quantities $V_{i\,j}=\mathrm{cov}\left(X_i,\,X_j\right)$ is called the covariance matrix.

The covariance obeys the identities

$$\mathrm{cov}\,(X+Z,\,Y)=\langle(X+Z)\,Y\rangle-\langle X+Z\rangle\,\langle Y\rangle \tag{13}$$

$$=\langle X\,Y\rangle+\langle Z\,Y\rangle-(\langle X\rangle+\langle Z\rangle)\,\langle Y\rangle \tag{14}$$

$$=\langle X\,Y\rangle-\langle X\rangle\,\langle Y\rangle+\langle Z\,Y\rangle-\langle Z\rangle\,\langle Y\rangle \tag{15}$$

$$=\mathrm{cov}\,(X,\,Y)+\mathrm{cov}\,(Z,\,Y). \tag{16}$$

By induction, it therefore follows that

$$\mathrm{cov}\left(\sum_{i=1}^{n}X_i,\,Y\right)=\sum_{i=1}^{n}\mathrm{cov}\,(X_i,\,Y) \tag{17}$$

$$\mathrm{cov}\left(\sum_{i=1}^{n}X_i,\,\sum_{j=1}^{m}Y_j\right)=\sum_{i=1}^{n}\mathrm{cov}\left(X_i,\,\sum_{j=1}^{m}Y_j\right) \tag{18}$$

$$=\sum_{i=1}^{n}\mathrm{cov}\left(\sum_{j=1}^{m}Y_j,\,X_i\right) \tag{19}$$

$$=\sum_{i=1}^{n}\sum_{j=1}^{m}\mathrm{cov}\left(Y_j,\,X_i\right) \tag{20}$$

$$=\sum_{i=1}^{n}\sum_{j=1}^{m}\mathrm{cov}\left(X_i,\,Y_j\right). \tag{21}$$

SEE ALSO

Bivariate Normal Distribution, Correlation Coefficient, Covariance Matrix, Statistical Correlation, Variance

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EXPLORE WITH WOLFRAM|ALPHA

WolframAlpha covariance (3,4,5,6,8) (1,1,1,1,1) =

More things to try: = covariance (3,4,5,6,8) (1,1,1,1,1) = AAPL WMT CSCO covariance matrix return

= covariance (165; 171; 168; 165; 176; 150; 164; 181; 172; 168; 175)

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